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UQ Fintech Society

# AlgoJam 3

## Instrument Specification

August 2026

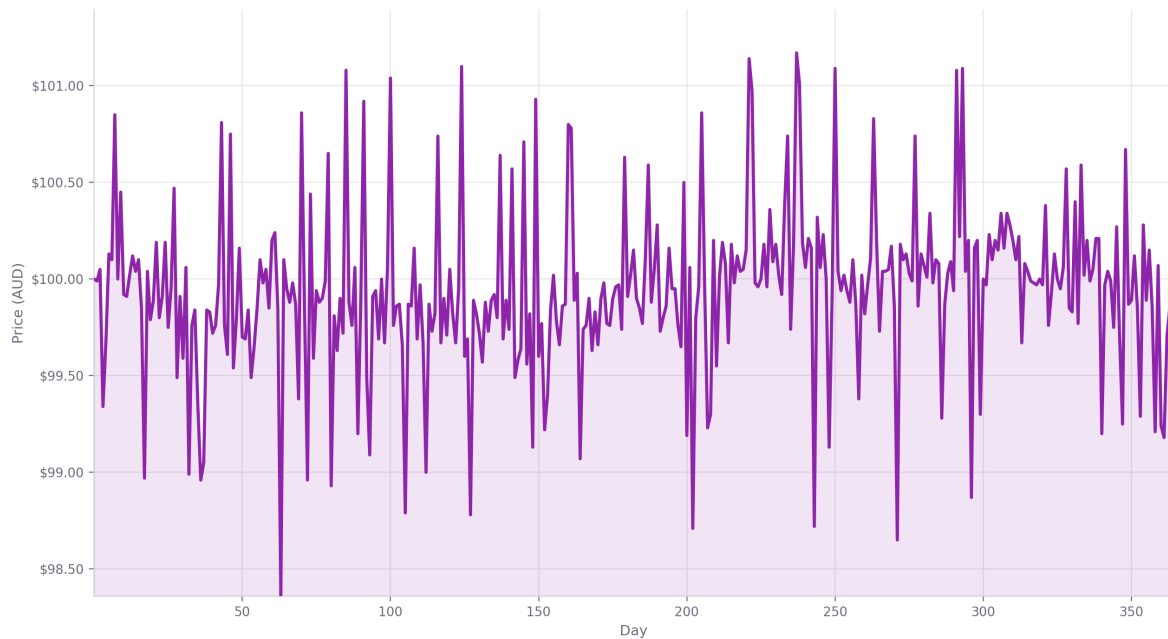
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### Introduction

Nine instruments have already traded through a simulated year, and your team's task is to write an algorithm that will trade them during the second year. All positions must be integers, each instrument has a position limit that applies equally to long and short positions, and the total portfolio value (the sum of the absolute value of each position multiplied by its price) must remain below **\$600,000 AUD** per day.

The charts below show only **Round 1 data**, which is the historical data provided for strategy development. Your algorithm is evaluated on Round 2, the following simulated year, which is revealed only on presentation night.

## UQ Dollar

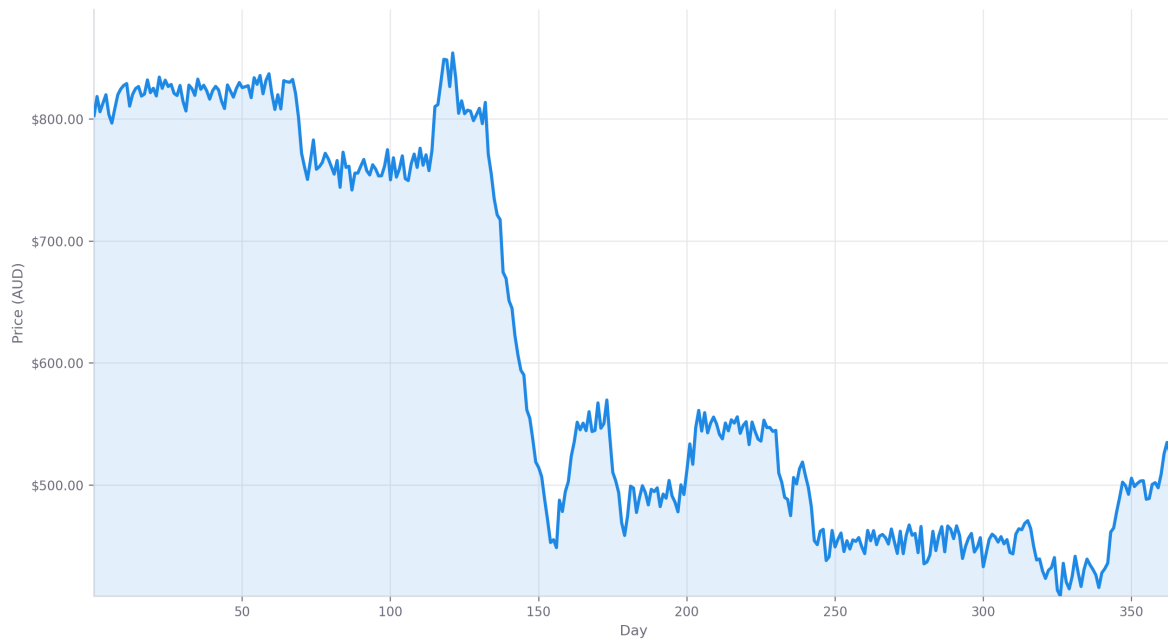


The UQ Union has started minting and maintaining their own currency, for use at UQ. It's pegged to the Australian Dollar at \$100, and serves as the primary medium for student transactions for food and drinks.

It might wander off during a busy week on campus and is usually back near a hundred a day or two later. The peg has held all year, in both directions where the up-spikes correct just as reliably as the down-dips.

**Position Limit:** 650 UQ Dollars

## Fintech Token



Made by three past-Fintech Execs who have since graduated and stopped replying to emails. Nobody has touched the code since. It sits still for weeks at a time, then someone posts something in the group chat and it jumps to a new price and stays there.

The stable and volatile periods have completely different shapes, so make use of more than one strategy to capture the two cases.

**Position Limit:** 100 Fintech Tokens

## Thrifty Jeans



From \$40 at an op-shop bin on a Sunday morning then resold at \$90 a year later, perhaps this is a sign of a new, steady, upwards trend for these old thrifty jeans?

**Position Limit:** 800 Pairs of Jeans

## Boat Party Ticket



Tickets sell out in about four minutes and then get resold in the group chats for whatever people will pay. Fun drinks are fizzin' and school is splashing every semester!

Demand follows the semester fairly closely, and nobody wants to be on a boat during exams. The same shape turns up at the same point every year, though the size of each peak isn't identical from one year to the next; the calendar is the reliable part, not the height.

**Position Limit:** 1,000 Tickets

## Sausage Sizzle



Bunning's lunch deal of a sizzle + drink over a folding table is a great way for to raise some money, but just remember to offer both tomato and BBQ sauce to play it safe.

The price is really just the ingredients added up: bread, sausages, and paying whoever is on the tongs for four hours. Also, whoever is on the roster does the books at the end of the day rather than the start, so today's price comes off yesterday's shopping.

Two of those three costs (bread and sausages) are tradable on the exchange; but, the third one is the volunteer - who, as it happens, also drives for MenuDash on weekends.

**Position Limit:** 3,000 Sausage Sizzles

## Bread



White, sliced, sold by the loaf.

Bread prices generally rise over time as flour becomes more expensive. Within a year, prices tend to increase gradually, but over the course of a week they often move in larger, less predictable jumps.

It also turns up on the sizzle's shopping list.

**Position Limit:** 500 Loaves

## Sausage



From the butcher on Hawken Drive, who has had the same weekend order for about eleven years.

Same shape as the bread: climbing slowly, noisy in the short run, and on the same shopping list.

**Position Limit:** 5,000 Sausages

## MenuDash



\$2.99 delivery, plus a service fee, plus a small-order fee, plus tax, plus a markup, plus one the app doesn't really explain.

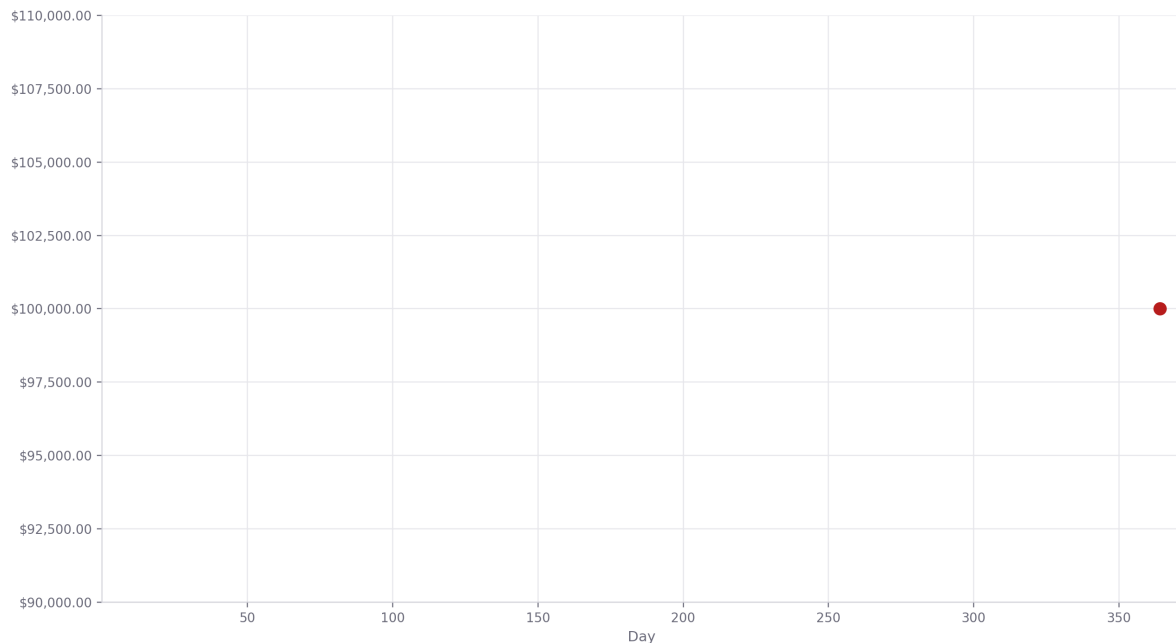
The fee isn't really about delivery. Most of the couriers are students, and a fair few of them spend their weekends behind a folding table in a hardware-store carpark; so, when their time becomes more valuable, the fee climbs with it. It has been climbing all year.

On top of that, the app rounds, surges and reshuffles its fees constantly, so any given day's number is a rough read on the thing underneath rather than an exact one. MenuDash posts a price every day. The cost driving it never does.

**Position Limit:** 75,000 Units

## Liferaft Ticket

Liferaft Ticket's price isn't pre-generated. It's computed **live**, during marking, from what every team decides. There is nothing to backtest against.



The ship is sinking! There's only one Liferaft ticket left. Every team must decide whether to buy it, sell it, or stay out.

The ticket starts at **\$100,000** on day one and has a position limit of  $\pm 1$ . The majority is worked out only from teams who actually take a side that day; staying flat means you're left out of the count entirely, not counted as a third option.

- If the majority of teams are **long**, the price **falls by \$5,000**.
- If the majority of teams are **short**, the price **rises by \$8,000**.
- If there is a tie, or every team remains flat, the price does not change.

Whichever vessel (liferaft or ship) the room crowds onto is the side that loses. There's no price history to fit; the only thing to model is everyone else. The price cannot fall below **\$20,000**. Remember, you will be able to access the historical price of this instrument as the year goes on.

**Position Limit:** 1 Ticket